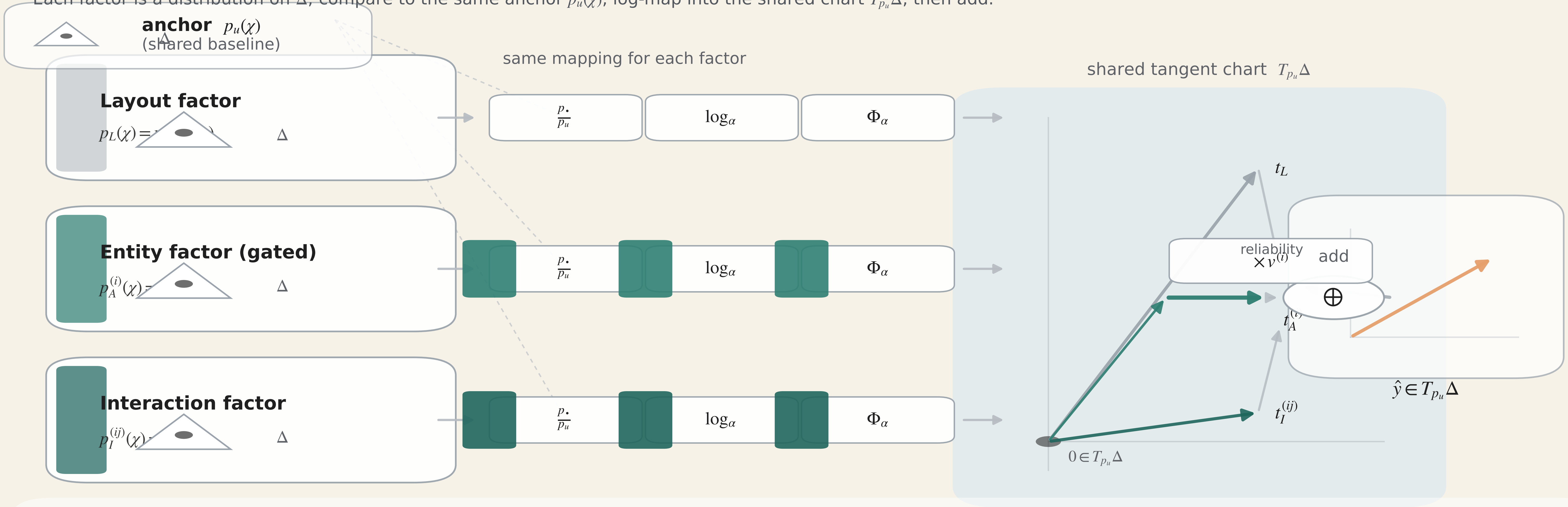
From distributions to tangent vectors (factor-wise)

Each factor is a distribution on Δ; compare to the same anchor $p_u(\chi)$, log-map into the shared chart $T_{p_u}\Delta$, then add.



$$t_\bullet = \Phi_\alpha\left(\log_\alpha \frac{p_\bullet}{p_u}\right) \in T_{p_u}\Delta, \quad \bullet \in \{L, A, I\}$$

$$\hat{y} = t_L + \sum_i v^{(i)} t_A^{(i)} + \sum_{i<j} t_I^{(ij)}$$

Bridge intuition: the log-map places all factor evidence into a common local chart at the same anchor $p_u(\chi)$, where additivity is valid; the gate $v^{(i)}$ acts as a reliability weight.